

MSE 211: Computational Methods

Lab 1 Report

Spring 2024

Group 5

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Abstract

This study explores the application of numerical methods to determine the behavior of a four-bar linkage mechanism, leveraging the insights obtained from graphical analysis to inform the selection of initial guesses for root-finding. The Bisection and Secant methods were employed to compute the angular positions of the mechanism over a complete rotation of the input link. The findings were corroborated using an interactive simulation tool to ensure the accuracy and consistency of the mechanism's operation within a single assembly mode. The research underlines the significance of integrating graphical methods with numerical algorithms for enhanced precision in mechanical design analysis.

Introduction

The investigation into the kinematics of four-bar linkage mechanisms is a fundamental aspect of mechanical design, facilitating the translation of motion and force within a variety of applications. The determination of link angles is critical for ensuring the desired motion path and is often challenged by the complexity of the system's behavior. In this context, the graphical method serves as an initial step, providing a visual approximation of potential roots indicative of angular positions. To refine these approximations, numerical methods such as the Bisection and Secant methods are introduced, offering iterative refinement towards precise solutions. This Lab report describes the integration of these methodologies, the rationale behind initial guess selections, and the verification process through a virtual simulation, highlighting the practical implications and advantages of such an approach.

Graphical Method

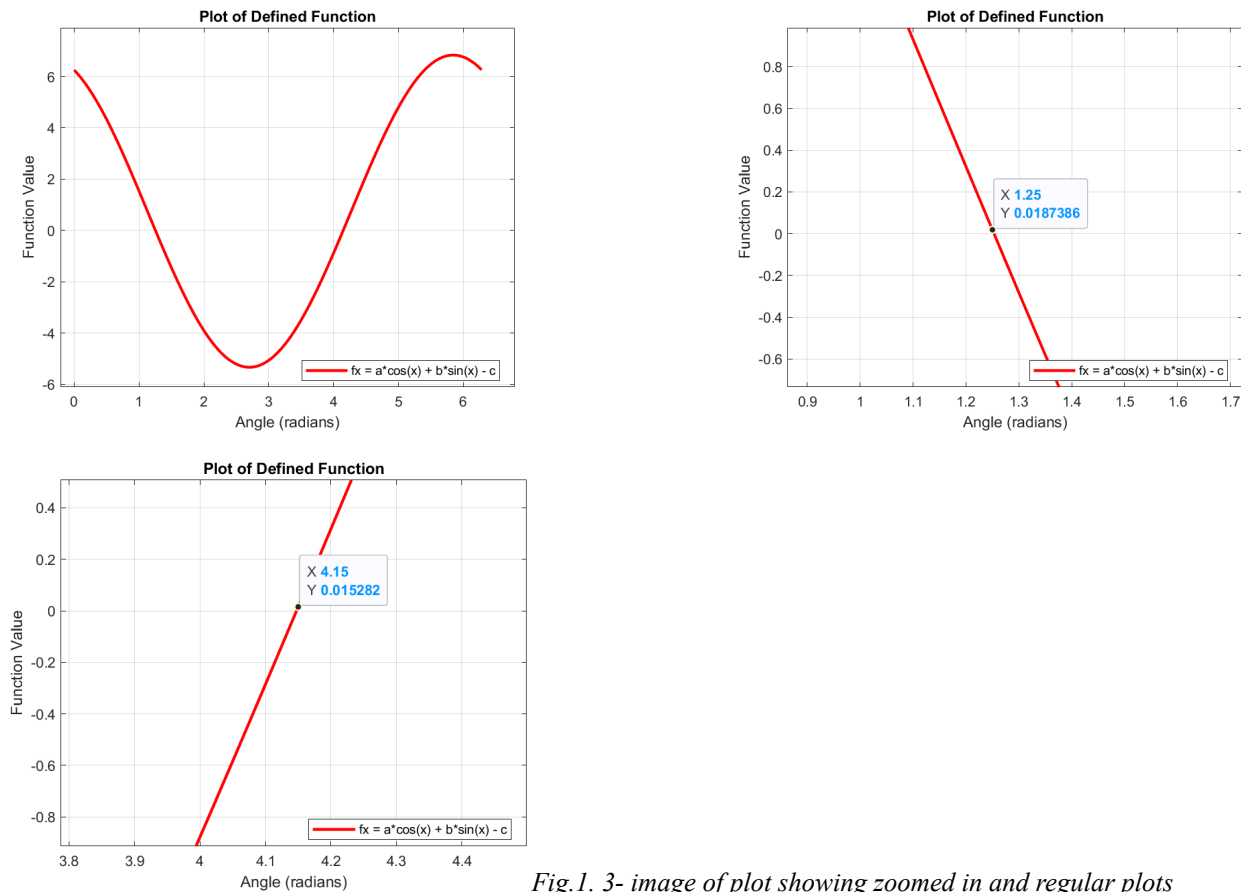


Fig.1. 3- image of plot showing zoomed in and regular plots

Guess within Specified range

Root 1 $\Rightarrow 1.24 < x_1 < 1.26$, Root 2 $\Rightarrow 4.14 < x_2 < 4.16$

Reasoning Behind Graphical method

The graphical method is a valuable first step in the root-finding process. It offers an intuitive and visual means to identify where the function crosses the x-axis, thereby indicating potential roots. This method is particularly useful for complex functions or when an initial guess is needed for further numerical analysis.

Bisection vs Secant method

Bisection Method

Command Window

1. First root in interval $0 < x < 2\pi$.
 $xL = 1.24$; % Lower bound
 $xU = 1.26$; % Upper bound

```
>> RunBisection
```

```
Root found: xr = 1.253104 after 14 iterations
```

```
f(xr) = 0.000000
```

2. Second root in interval $0 < x < 2\pi$.

```
xL = 4.14; % Lower bound
```

```
xU = 4.16; % Upper bound
```

```
>> RunBisection
```

```
Root found: xr = 4.147469 after 14 iterations
```

```
f(xr) = 0.000001
```

Secant Method

Command Window

1. First root with guess bracketing the estimate point

```
>> x0 = 1.24; % Initial estimate 1
```

```
x1 = 1.26; % Initial estimate 2
```

```
err = 1e-6; % Tolerance error
```

```
[xr, fxr] = Secant(x0, x1, err);
```

```
Root found: xr = 1.253104 after 3 iterations, f(xr) = -0.000000
```

2. Second root guess without bracketing the estimate point

```
>> x0 = 5; % Initial estimate 1
```

```
x1 = 6; % Initial estimate 2
```

```
err = 1e-6; % Tolerance error
```

```
[xr, fxr] = Secant(x0, x1, err);
```

```
Root found: xr = 4.147469 after 7 iterations, f(xr) = 0.000000
```

The Bisection and Secant methods are both numerical techniques used to find roots of a function, where a "root" is a solution to the equation:

$$f(x)=0 \text{ [function whose roots gave value of Angle]}$$

The choice of method and initial points significantly affects the speed, accuracy, and even the success of the root-finding process. Understanding the usage of roots obtained graphically can provide valuable insights into these methods and their outcomes.

Usage of Graphically Obtained Roots

Graphically obtained roots give a visual approximation of where the function crosses the x-axis. These approximations are beneficial for:

1. *Selecting Initial Points for the Secant Method:*

The Secant method requires two initial guesses. Graphical estimations can help ensure these guesses are close to an actual root, potentially leading to faster convergence.

2. Setting Bounds for the Bisection Method:

The Bisection method requires a bracket $[a, b]$ within which the root lies (i.e., $f(a)$ and $f(b)$ have opposite signs). Graphical methods can identify such intervals.

Results Using Numerical Methods

Bisection Method:

It converges to a root with guaranteed accuracy if the function is continuous within the interval and $f(a)$ and $f(b)$ have opposite signs. The method is relatively slow compared to the Secant method, as it linearly narrows down the search interval. However, its reliability and ease of implementation are significant advantages.

Secant Method:

It offers faster convergence than the Bisection method since it uses a form of the derivative to estimate the next point. However, its convergence is not guaranteed if the initial points are not chosen wisely. The method can fail or converge to an unexpected root if the initial guesses are far from the true root.

Speed and Accuracy

Speed:

The Secant method generally converges faster than the Bisection method because it uses a more aggressive approach to find the root, utilizing the slope of the line connecting two points on the function. However, this speed comes at the cost of potentially overshooting or diverging if the initial points are not well-chosen.

Accuracy:

The Bisection method provides steady, predictable convergence, making it highly reliable for achieving desired accuracy, albeit slowly. The Secant method, while faster, can sometimes offer less predictable accuracy, especially with poorly chosen initial points, particularly when secant line becomes parallel to the x-axis.

Impact of Initial Points

Initial Points from Graphical Estimations:

Using initial points close to the actual root (as suggested by graphical analysis) tends to make both methods more effective. For the Bisection method, it ensures the root lies within the initial bracket, reducing the number of iterations. For the Secant method, closer initial points generally mean faster convergence to the correct root.

Initial Points Far from Graphical Estimations:

Deviating from graphically suggested points can severely impact the Secant method, leading to convergence to the wrong root or divergence. The Bisection method is less sensitive but might require more iterations if the initial bracket is large.

Implementation

Verification

This tool allows to visualize the movement and ensure there are no assembly mode jumps as θ_2 completes a 360-degree rotation. The visual comparison can validate the continuity and range of θ_3 and θ_4 as expected from the calculations. The website provides an interactive model to set the link lengths and observe the resulting motion, which is a practical way to cross-check your theoretical results.

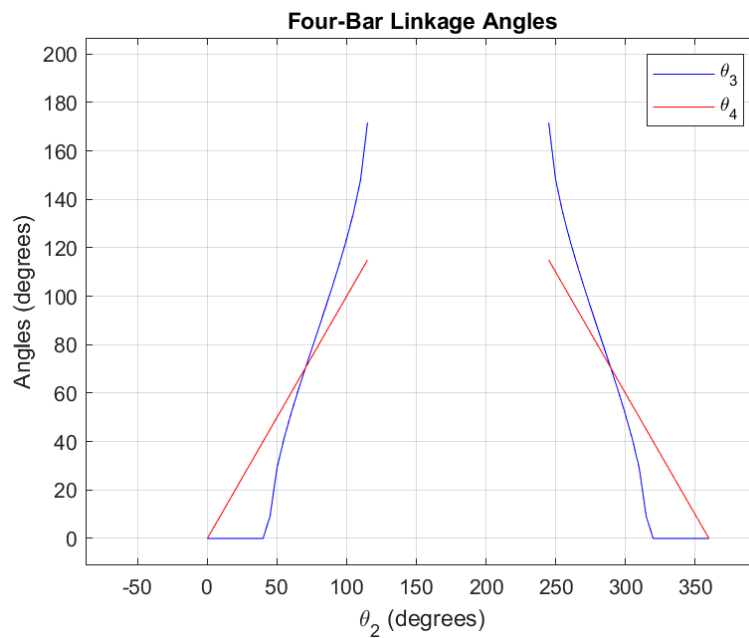
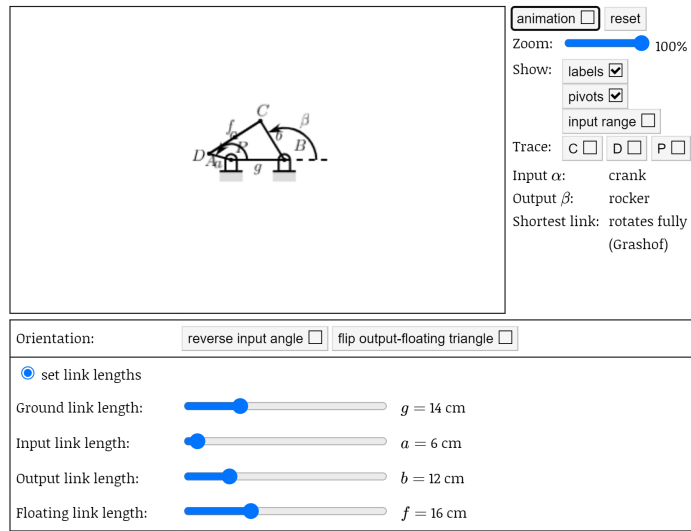


Fig.2. 4-bar linkage mechanism

The acceptable range for θ_3 and θ_4 in a four-bar linkage mechanism, considering only one assembly mode and knowing that θ_2 is rotating 360 degrees, is determined by the physical constraints of the mechanism and the geometry of the linkages.

1. Physical Constraints: Recognize that in real-life applications, the linkages cannot overlap or intersect in ways that are physically impossible. The range of motion for θ_3 and θ_4 is limited by the lengths of the links and their starting positions.

2. Geometry of the Linkages: Based on the loop closure equation and the cosine rule, determine the theoretical limits for θ_3 and θ_4 without considering physical interferences. These limits are based on the mathematical solutions to the equations for any given θ_2 .

When θ_2 rotates 360 degrees, theoretically, the mechanism may pass through positions that correspond to a different assembly mode (also known as circuit or branch change). However, physically, the mechanism cannot jump between these modes without disassembly. Therefore, the acceptable range for θ_3 and θ_4 should be continuous and smooth throughout the rotation, without sudden jumps in angle values that would indicate a change in assembly mode.

Conclusion

The graphical method serves as a crucial preliminary step in understanding the behavior of a function and identifying potential roots. It is especially advantageous as a visual tool, providing intuitive insights that guide subsequent numerical methods.

For the given four-bar linkage mechanism, graphical analysis suggested initial guesses for roots within specific ranges. These guesses played a foundational role in applying the Bisection and Secant methods, which were then employed to find the roots with high precision.

The Bisection method, while slower, offers a more controlled and reliable approach to converge upon the root, making it highly suitable for scenarios where certainty in finding a root is paramount. On the other hand, the Secant method is faster but relies heavily on the proximity of initial estimates to the true root for effective convergence. It can sometimes yield less predictable results if the initial guesses are not close to the actual root.

The results indicate that, with carefully chosen initial guesses from graphical methods, both the Bisection and Secant methods successfully found the roots within the intervals specified, with the Secant method requiring fewer iterations. However, the Bisection method's reliability might make it preferable in applications demanding guaranteed convergence.

In conclusion, the combined use of graphical analysis and numerical methods offers a robust strategy for identifying and verifying the roots of functions. The verification process, supported by interactive tools such as the one provided by the University of Illinois Urbana-Champaign's

Dynamics website, allows for visual validation and ensures that the mechanisms behave as theoretically predicted without shifting between assembly modes.

Included M.files: funct, bisection, secant, implementation, SecantMethodForTheta, RunSecant